



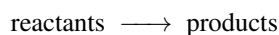
Ch. 4. Chemical equilibrium

REACTIONS rarely occur to completion. Once products form, they can follow a backward reaction to regenerate the reactants, and at the same time, reactants regenerate and produce products. Overall, chemical reactions proceed until reaching equilibrium. The equilibrium conditions determine how profitable a reaction can be as they describe how much of the products stay as products. This chapter covers the idea and basic principles of chemical equilibrium. We will describe the role of an equilibrium constant, which gives insight into the mixtures of reactants and products in equilibrium. Also, we will discuss the different types of equilibrium constants in terms of molarity and pressure and the role of a reaction quotient that gives insight into the direction in which a chemical reaction proceeds towards the equilibrium. Finally, we will cover Le Châtelier principle that describes the direction in which a chemical reaction proceeds to recover the equilibrium state, after equilibrium is altered.

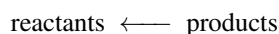
4.1 Chemical Equilibrium

This section covers basic ideas about chemical equilibrium. First, we will introduce the concept of the forward and reverse reactions. Next, we will define the idea of equilibrium, based on the speed of the forward and reverse reactions.

Forward and reverse reactions In chemical reactions, reactants form products. We call this the *forward reaction*:



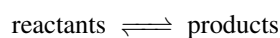
However, once the products form they can also generate reactants through the *reverse reaction*:



Equilibrium in terms of rate In chemical reactions, both the forward and the reverse reactions happen synchronously, so that when products form, reactants form as well. However, both the reverse (r_{reverse}) and forward (r_{forward}) reactions have different speeds. In the beginning, the forward reaction proceeds at a faster pace than the reverse so that the reaction advances. Once products form, the reverse reaction will start speeding up. A reaction reaches *chemical equilibrium* when both the forward and reverse reactions proceed at the same speed (see Figure 4.2).

$$r_{\text{forward}} = r_{\text{reverse}}$$

The term equilibrium in science refers to something remaining constant with time. Chemical reactions are normally written down with a double arrow that indicates equilibrium:



For example, the chemical process that forms ammonia from nitrogen and hydrogen can proceed using the forward and reverse reactions



Chemical equilibrium is a dynamic process. Even when the concentrations of reactants and products do not change with time, the reverse and forward reactions occur, and molecules still change at a microscopic level.

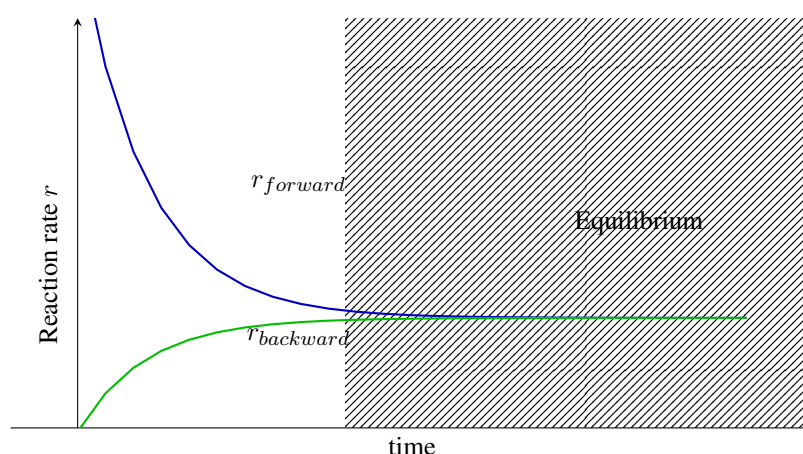


Figure 4.2 Changes with time of the rates for the forward and backward reaction, showing the point when equilibrium is reached.

Equilibrium in terms of concentration It is just a matter of time before a reaction reaches equilibrium. Initially, the forward process normally proceeds at high speed as products start to form. This is because initially there is an abundance of reactants. Once the products start to accumulate the forward process will start happening. This is because at that point there is an increasing amount of product molecules that can go back to reactants. Eventually, the forward and reverse rates become equal as the reaction reaches equilibrium. When the equilibrium has been reached, reactants and products have very specific concentrations that depend on temperature and are constant with time (see Figure 4.2). At the same time, a reaction in equilibrium contains a mixture of reactants and products.

▼ The forward reaction goes from reactants to products whereas the reverse reaction goes from products to reactants.

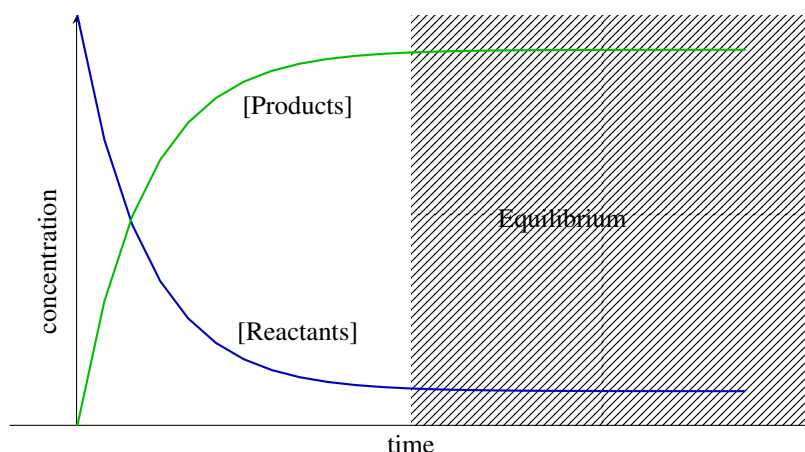
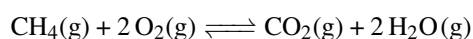


Figure 4.2 Change of the concentration of reactants and products with time showing the point when equilibrium is reached.

Sample Problem 1

Write down the forward and reverse reactions for:

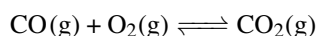


SOLUTION

(a) The forward reaction is $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$ and the reverse $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longleftarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$

STUDY CHECK

Write down the forward and reverse reactions for the reaction:



► Answer: $\text{CO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$ and $\text{CO}(\text{g}) + \text{O}_2(\text{g}) \longleftarrow \text{CO}_2(\text{g})$

4.2 The equilibrium constant

Reactions in equilibrium are characterized by an equilibrium constant. This next section will show how to interpret and calculate the value of the equilibrium constant of a reaction.

Equilibrium mixtures Imagine you start a chemical reaction. Initially, products will form while reactants disappear. At equilibrium, you will have a mixture of reactants and products with the forward and reverse processes happening at the same rate. Would an equilibrium mixture contain more reactants or more products? The equilibrium constant helps predict just that.

Table 4.1 Different K_c values at 298K

Reaction					K_c	Equilibrium mixture
$2 \text{NH}_3(\text{g})$	$\rightleftharpoons$	$\text{N}_2(\text{g})$	+	$3 \text{H}_2(\text{g})$	17	Products > Reactants
$\text{H}_2(\text{g})$ + $\text{I}_2(\text{g})$	$\rightleftharpoons$	$2 \text{HI}(\text{g})$			50	Products > Reactants
$2 \text{SO}_3(\text{g})$	$\rightleftharpoons$	$2 \text{SO}_2(\text{g})$	+	$\text{O}_2(\text{g})$	0.3	Products < Reactants
$\text{H}_2\text{O}(\text{l})$	$\rightleftharpoons$	$\text{H}_2\text{O}(\text{g})$			0.2	Products < Reactants

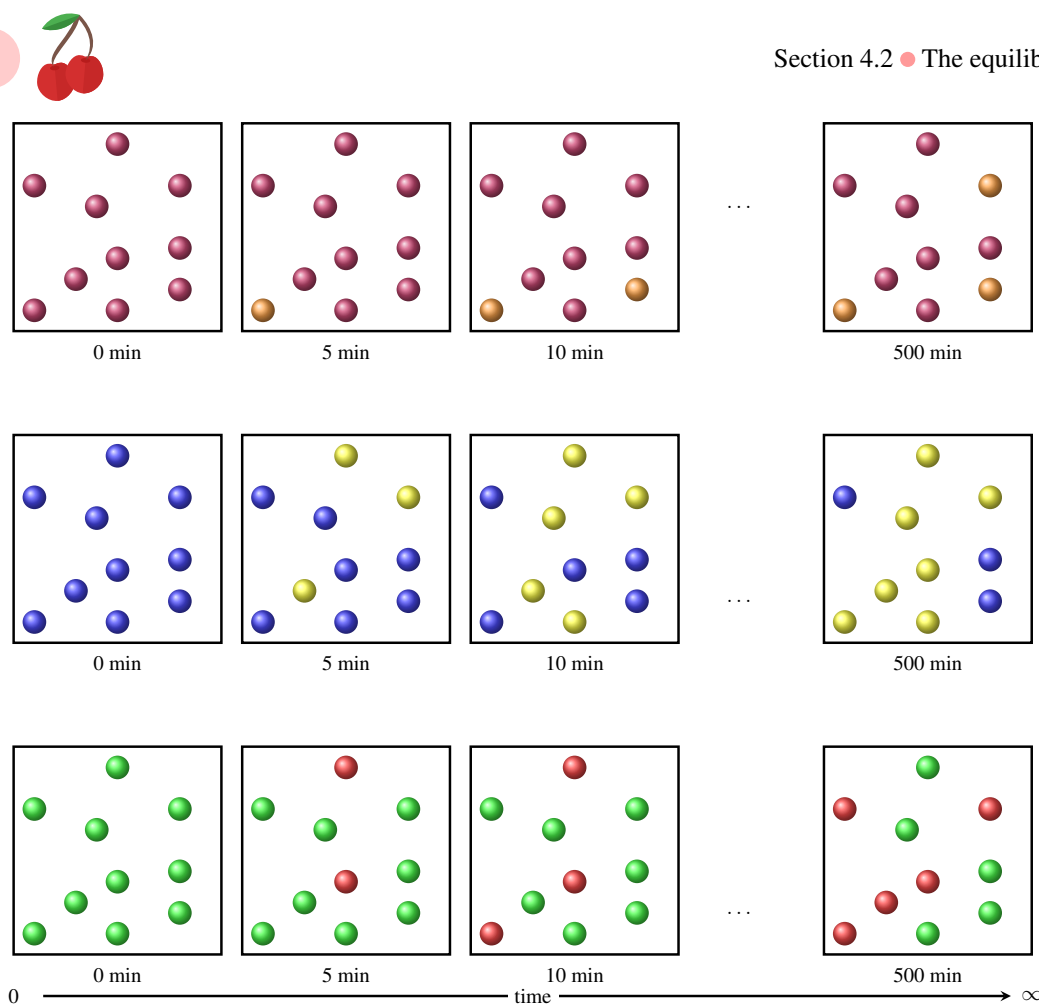
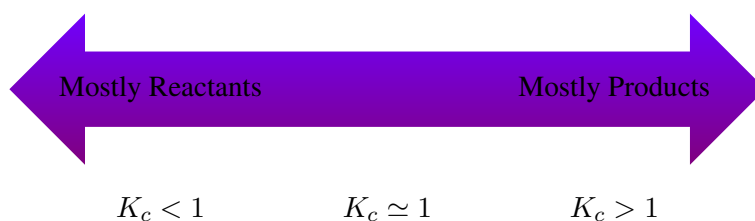


Figure 4.3 Three equilibrium situations. (Top Panel) At equilibrium, there are more reactants than products. (Central Panel) At equilibrium, there are more products than reactants. (Bottom Panel) At equilibrium, there are as many products as reactants.

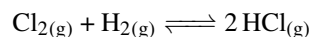
The equilibrium constant of a reaction The equilibrium constant associated with a reaction K_c indicates whether reactants or products are more abundant at equilibrium. Each reaction has one K_c value that depends only on temperature and the subscript c represents concentration. On one hand, if K_c is larger than one there will be a larger concentration of products than reactants in the equilibrium mixture (see Figure 4.3). On the other hand, if K_c is smaller than one there will be a larger concentration of reactants than products in the equilibrium mixture. If K_c is close to one then both reactants and products will have the same concentration in the equilibrium mixture (see Table 4.1 for examples).



In terms of units, equilibrium constants are unitless numbers—a number without units. In other words, they have no unit and they are simply expressed as a number.

Sample Problem 2

The value of K_c for the following reaction at 300K is 4×10^{31} . Indicate whether the equilibrium mixture will contain mostly reactants, mostly products or both.

**SOLUTION**

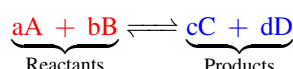
As K_c is larger than one, an equilibrium mixture of Cl_2 , H_2 and HCl will contain mainly products, that is will be mainly made of HCl .

STUDY CHECK

The value of K_c for $\text{F}_{2(g)} \rightleftharpoons 2 \text{F}_{(g)}$ at 500K is 7×10^{-13} . Indicate whether the equilibrium mixture will contain mostly reactants, mostly products or both.

► Answer: mostly reactants.

Equilibrium constant expression Let's consider a general equilibrium reaction in which A and B react to form C and D



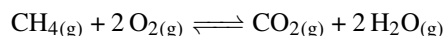
The stoichiometric coefficients of the reaction are a , b , c , and d . Cato Maximilian Gulberg and Peter Waage proposed the *law of mass action* in 1864 in order to describe a chemical system in equilibrium. The expression for the equilibrium constant, which defines the law of mass action, depends only on the concentration of the reactants and products at fixed temperature. To refer to concentration we will use square brackets. For example, $[A]$ means the concentration of A . Hence, the expression of the equilibrium constant will be:

$$K_c = \frac{[\text{Products}]}{[\text{Reactants}]} = \frac{[C]^c \cdot [D]^d}{[A]^a \cdot [B]^b} \quad \text{Equilibrium constant} \quad (4.1)$$

Let us break down the expression of K_c . On top of the fraction, we have the equilibrium concentration of the products to the power of its coefficients. For example $[C]^c$ means the equilibrium concentration of C to the power of the coefficient c . On the bottom of the fraction, we have the concentration of the reactants to the power of its coefficients. All concentrations in K_c are timed.

$$K_c = \frac{\overset{\text{product concentration}}{[C]^c \cdot [D]^d}}{\underset{\text{reactant coefficient}}{[A]^a \cdot [B]^b}}$$

Let's focus on the reaction below:



The expression of the equilibrium constant would be:

$$K_c = \frac{[\text{CO}_2] \cdot [\text{H}_2\text{O}]^2}{[\text{CH}_4] \cdot [\text{O}_2]^2}$$

As a final note, the equilibrium constant is a dimensionless property. Without going into too much detail, this is because on its more general definition, equilibrium constants

$K_c < 1$

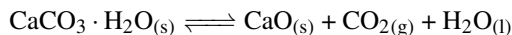
$K_c = 1$

$K_c > 1$

▼ The magnitude of the equilibrium constant indicates whether there is more products or reactants in an equilibrium mixture.

are defined in terms of *chemical activities*, a general dimensionless property that relates to amount of substance (for example, can be concentration for solutions, pressures for gases).

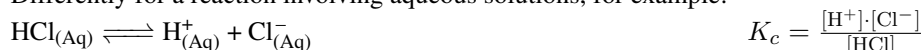
Equilibrium involving solids, liquids and solutions Let us analyze an example of a reaction involving solids or liquids:



Solids and liquids have no concentration and hence they should not be included in the expression of K_c . For the example above:

$$K_c = [\text{CO}_2]$$

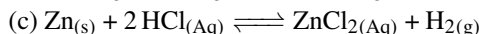
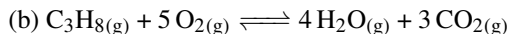
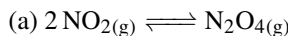
Differently for a reaction involving aqueous solutions, for example:



Overall, remember that in the expression of K_c , you can only include gases (g) or aqueous solutions (aq) as you can ignore solids and liquids without a well-defined concentration.

Sample Problem 3

Write down the expression of K_c for the following reactions:



SOLUTION

Remember you can only include gas and aqueous solution in the expression of K_c . For the first example

$$K_c = \frac{[\text{N}_2\text{O}_4]}{[\text{NO}_2]^2}$$

For the second example,

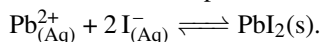
$$K_c = \frac{[\text{H}_2\text{O}]^4 \cdot [\text{CO}_2]^3}{[\text{C}_3\text{H}_8] \cdot [\text{O}_2]^5}$$

For the last example:

$$K_c = \frac{[\text{ZnCl}_2] \cdot [\text{H}_2]}{[\text{HCl}]^2}$$

STUDY CHECK

Write down the expression of K_c for the following reaction:



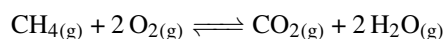
►Answer: $\frac{1}{[\text{Pb}^{2+}] \cdot [\text{I}^-]^2}$

Equilibrium constant in terms of pressures The equilibrium constant K_c is the ratio of the concentration of products and reactants. As concentration is related to pressure—remember the chapter about gases—we can also express an equilibrium constant in terms of pressures, we call this K_p .

$$K_p = \frac{P_{\text{Products}}}{P_{\text{Reactants}}} = \frac{P_C^c \cdot P_D^d}{P_A^a \cdot P_B^b} \quad \text{Equilibrium constant} \quad (4.2)$$



Let's focus on an example. Think about the reaction below that involved just gases:



The expression of the equilibrium constant would be:

$$K_p = \frac{P_{\text{CO}_2} \cdot P_{\text{H}_2\text{O}}^2}{P_{\text{CH}_4} \cdot P_{\text{O}_2}^2}$$

K_p values larger than one implies that in the mixture the pressure of products is larger than the pressure of reactants and the opposite is true for values smaller than one.

Relating K_c and K_p The values of K_c and K_p are related by the following formula:

$$K_p = K_c(RT)^{\Delta n} \quad (4.3)$$

where:

K_p is the equilibrium constant in terms of pressure

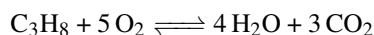
K_c is the equilibrium constant in terms of concentration

R is the constant of the gases in pressure units (0.082 atmL/molK)

T is the absolute temperature (in Kelvins)

Δn is the change of stoichiometry of the reaction

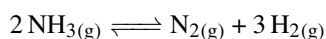
Let us analyze the role of Δn for the reaction:



Δn represents the number of molecules of products with respect to the number of molecules of reactants: $7-6=1$. For this reaction, we have that $K_p = K_c(RT)$. However, the power of the relationship will depend on the stoichiometry of the reaction. At the same time, the relationship between both constants depends on the temperature.

Sample Problem 4

For the following reaction



The value of K_c at 300K is 17. Calculate the value of K_p at the same temperature.

SOLUTION

First we will write down the expression of both equilibrium constants:

$$K_c = \frac{[\text{N}_2] \cdot [\text{H}_2]^3}{[\text{NH}_3]^2} \quad K_p = \frac{P_{\text{N}_2} \cdot P_{\text{H}_2}^3}{P_{\text{NH}_3}^2}$$

We will also calculate Δn in order to establish the relationship between both constants:

$$\Delta n = 1 + 3 - 2 = 2$$

We have that:

$$K_p = K_c(RT)^{\Delta n} = K_c(RT)^2$$

Given that K_c at 300K is 17:

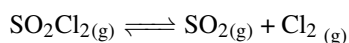
$$K_p = K_c(RT)^{\Delta n} = 17(0.082 \cdot 300)^2 = 10287$$



In equilibrium, the pressure of products will be larger than the pressure of reactants.

◆ STUDY CHECK

For the following reaction



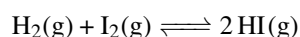
The value of K_p at 300K is 0.05. Calculate the value of K_c at the same temperature.

►Answer: 2×10^{-3}

4.3 Using equilibrium constants

We saw that the equilibrium constant of a reaction tells you whether there are more reactants or products in an equilibrium mixture. At the same time, one can use K_c to quantitatively calculate the value of the equilibrium concentration of reactants and products. This section will explain how to do this.

Solving from K_c Let's analyze the reaction of hydrogen (H_2) and iodine (I_2) to produce hydrogen iodide (HI):



The equilibrium constant at 300K is $3 \cdot 10^{-1}$. Analyzing a reaction mixture we find that the concentration of H_2 is 1M and the concentration of I_2 is 2M. We want to calculate how much HI we have in the mixture. As the concentrations of reactants and products are linked together through K_c we can certainly solve for $[\text{HI}]$. The expression for K_c is:

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2] \cdot [\text{I}_2]}$$

and we know that $[\text{H}_2] = 1M$ and $[\text{I}_2] = 2M$. Plugging the values in the expression of K_c , and given the numerical value of K_c we have:

$$3 \cdot 10^{-1} = \frac{[\text{HI}]^2}{1 \cdot 2}$$

Solving for $[\text{HI}]$ we have

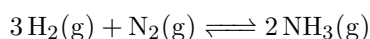
$$[\text{HI}]^2 = 0.6$$

To solve for $[\text{HI}]$ we have to use an square root:

$$[\text{HI}] = \sqrt[2]{0.6} = 0.77M$$

Sample Problem 5

The value of the equilibrium constant for the reaction





is $3 \cdot 10^8$ at 300K. An analysis of an equilibrium mixture gave a concentration of nitrogen and ammonia of 2M, respectively. Calculate the equilibrium concentration of hydrogen at 300K.

SOLUTION

The value of the equilibrium constant for the formation of ammonia is:

$$K_c = \frac{[\text{NH}_3]^2}{[\text{H}_2]^3 \cdot [\text{N}_2]}$$

We know $[\text{NH}_3]$ and $[\text{N}_2]$ and both values are 2M, and we also know $K_c = 3 \cdot 10^8$. Plugging these values into the previous equations we obtain:

$$3 \cdot 10^8 = \frac{2^2}{[\text{H}_2]^3 \cdot 2}$$

we can solve for $[\text{H}_2]$:

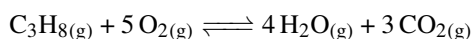
$$[\text{H}_2]^3 = 6.6 \cdot 10^{-9}$$

In order to obtain $[\text{H}_2]$ we need a cubic root:

$$[\text{H}_2] = \sqrt[3]{6.6 \cdot 10^{-9}} = 1.9 \cdot 10^{-3} \text{M}$$

STUDY CHECK

The value of the equilibrium constant for the reaction



is 500 at a given temperature. An analysis of an equilibrium mixture gave a concentration of water, carbon dioxide and C_3H_8 of 1M. Calculate the equilibrium concentration of oxygen at that temperature.

►Answer: $[\text{O}_2] = 0.29 \text{M}$

4.4 Concentration or pressure ratio

In a chemical reaction, if we prepare a mixture of reactants they will generate products. At equilibrium reactants and products exist at a very specific ratio given by the equilibrium constant. What if we would prepare a mixture containing both reactants and products? Given that both reactants and products are related by an equilibrium, how would the concentrations change? In other words, would the reaction proceed toward the right, hence producing products, or the left to produce reactants? This section introduces the use of a concentration (or pressure) ratio that helps predict the direction in which a mixture of reactants and products will proceed toward equilibrium.

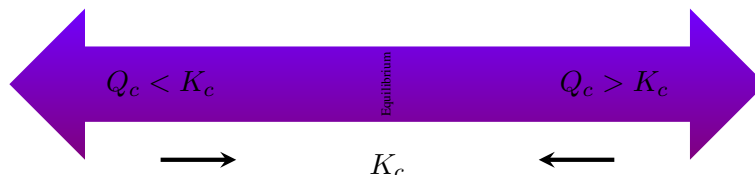
Definition of concentration ratio Equilibrium constants are ratios of equilibrium concentration or pressure of products over reactants measured after the system has reacted for an infinite time. Concentration ratios Q_c are the ratio of the concentration of products over reactants away from the equilibrium. In other words,



they represent the reaction away from the equilibrium.

$$Q_c = \frac{[\text{Products}]_{\text{noneq}}}{[\text{Reactants}]_{\text{noneq}}} = \frac{[C]_{\text{noneq}}^c \cdot [D]_{\text{noneq}}^d}{[A]_{\text{noneq}}^a \cdot [B]_{\text{noneq}}^b} \quad \text{concentration ratio} \quad (4.4)$$

Use of concentration ratios Concentration ratios help predict whether a reaction will proceed toward the left or toward the right when we start with a mixture of reactants and products away from equilibrium.



If Q_c is larger than K_c the reaction will proceed towards the left producing reactants. On the other hand, if Q_c is smaller than K_c the reaction will proceed towards the right producing products. If Q_c equals K_c then the reaction is in equilibrium.

Sample Problem 6

In two different experiments, we prepare mixtures of four chemicals (molarities indicated in the table) interconnected by the following equilibrium:



Indicate if any of the mixtures is in equilibrium. If it is not, indicate whether the reaction will evolve towards the left or the right to reach equilibrium.

Experiment	CO _(g)	+	H ₂ O _(g)	⇌	CO _{2(g)}	+	H _{2(g)}
1	0.0203		0.0203		0.0040		0.0040
2	0.011		0.0011		0.037		0.046

SOLUTION

We will compute the concentration ratio for each experiment and compare the value with k_c . If Q_c is larger than k_c the reaction will proceed to the left whereas if Q_c is smaller than k_c the reaction will proceed to the right.

Experiment	CO _(g)	+	H ₂ O _(g)	⇌	CO _{2(g)}	+	H _{2(g)}	Q_c
1	0.0203		0.0203		0.0040		0.0040	3.8×10^{-4}
2	0.011		0.0011		0.037		0.046	140

None of the mixtures are in equilibrium as Q_c differs from k_c . In the first experiment the reaction will proceed to the right and in the second experiment the reaction will proceed to the left.

STUDY CHECK

We prepare mixtures of four chemicals (molarities indicated in the table) in an experiment. These chemicals are interconnected by the following equilibrium:



Indicate if the mixture is in equilibrium. If it is not, indicate whether the reaction will evolve towards the left or the right to reach equilibrium.

Experiment	CO _{2(g)}	+	H ₂ O _{2(g)}	⇌	CO _{2(g)}	+	H _{2(g)}
1	0.0094		0.0025		0.0005		0.0076

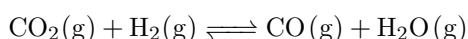
►Answer: to the right

4.5 Le Chatelier principle

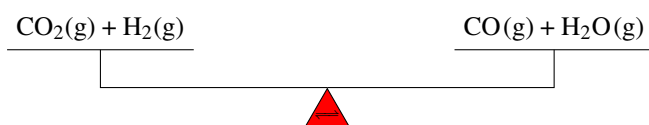
At this point, we have covered the idea of equilibrium and we have seen that the forward and reverse reactions have the same rate at equilibrium. Now, what happens if you alter this equilibrium? Le Châtelier principle claims a reaction will go back to its original equilibrium state by shifting left or right.

Le Chatelier principle When a reaction is in equilibrium the forward and reverse reactions proceed at the same speed. Also in an equilibrium state, the concentrations of reactants and products have very specific values. Imagine that you create stress conditions by adding reactants or products or even changing the temperature. This stress will have an impact on the equilibrium and the reaction eventually will reach a new state of equilibrium by somehow counteracting this stress. Le Châtelier principle states that when stress is placed in a reaction (adding or removing reactant or products, increasing or decreasing temperature) the equilibrium will be shifted in the direction that relieves that stress. Table 4.2 displays different aspects regarding Le Châtelier's principle in terms of parameter change and consequence.

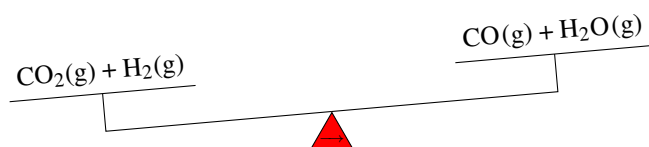
Change in concentration Let us consider the following equilibrium:



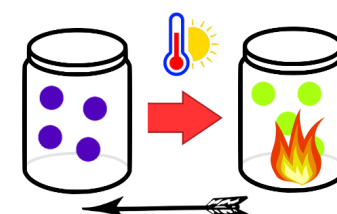
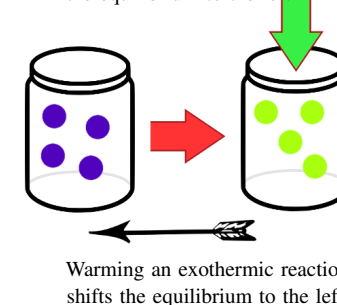
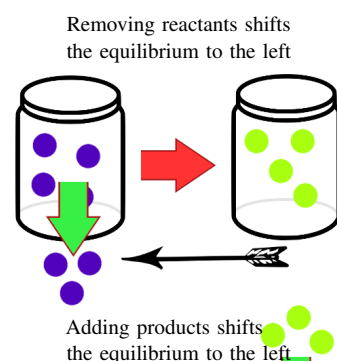
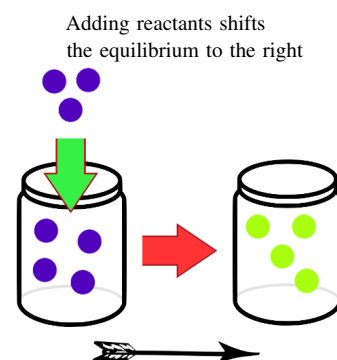
K_c for this reaction equals one at 1200K. This means that the concentration of reactants and products are the same. We can represent this using this balance or seesaw



If we add some CO₂ the equilibrium will be affected. To counteract this stress, the reaction will restore the equilibrium by decreasing the amount of CO₂. This can only be achieved by displacing the equilibrium to the right so that CO₂ is removed. Mind that CO₂ is consumed if the reaction moves from reactants → to products and it is produced when going from products ← to reactants. We can represent this with the following seesaw.



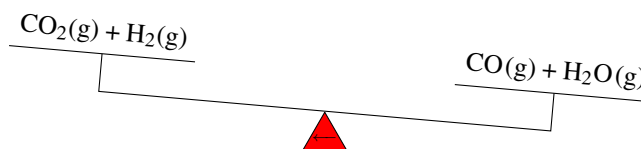
▼Henry Louis Le Châtelier was a French chemist who devised a principle used by chemists to predict the effect a changing condition has on a system in chemical equilibrium.



▼Le Châtelier principle helps predict the outcome of altering an equilibrium mixture.



As we added CO_2 the reactants now weigh more and hence the reaction has to proceed to the right $\longrightarrow$. Now imagine we remove some $\text{CO}_2(\text{g})$. Again, the equilibrium will be affected and the reaction will restore its equilibrium state by doing the opposite, that is producing $\text{CO}_2(\text{g})$ as the reaction proceeds from reactants $\longleftarrow$ to products. Again using the seesaw:

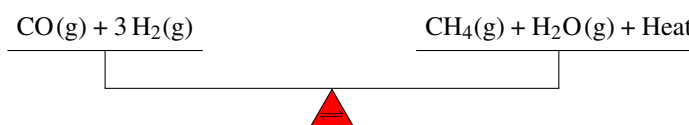


We can also add a different chemical that is not involved in the equilibrium. In this case, the equilibrium will not be affected by this change.

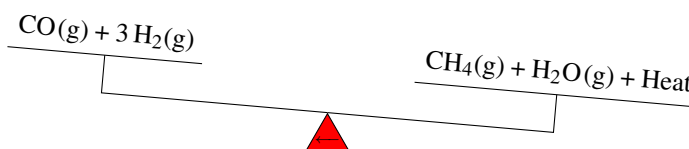
Temperature change Let us consider the following equilibrium that produces heat—remember we describe these types of reactions as exothermic:



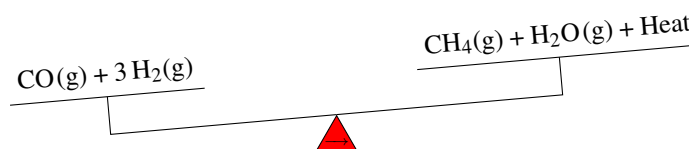
Again, this reaction is in equilibrium so we can use the same seesaw analogy.



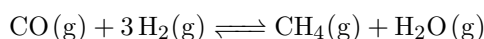
If we increase the temperature of the system, the equilibrium will be affected. To go back to an equilibrium state the reaction will decrease the temperature of the container. As the reaction produces heat, a way to decrease the system temperature is to generate reactants ($\longleftarrow$). Again, using the scale that means:



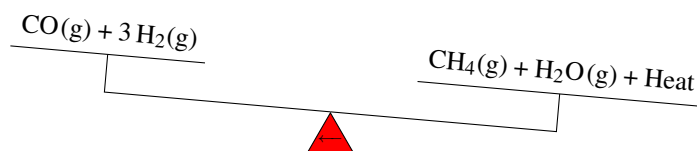
Differently now if we decrease the temperature, the reaction will increase the temperature by going back to its equilibrium state going from reactants $\longrightarrow$ to products. This is because heat is produced as a product



Volume change We can also think about increasing or decreasing the volume in which the reaction takes place. This change will have an impact on the reaction equilibrium as the concentrations of reactants and products will be altered by this change. Changes in volume will shift the reaction towards the left or right depending on the overall stoichiometric change of the reaction, that is on whether the reaction produces or consumes molecules. For reactions that generate matter, that is, in the case that $\Delta n > 0$, increasing the volume will follow the increase of the number of moles. In other words, by increasing the volume, the equilibrium will shift towards the products, that is towards the right. For reactions that consume matter ($\Delta n < 0$), increasing the volume will shift the equilibrium towards the reactants, that is towards the left. For example, the reaction below consumes molecules:



if we increase the volume of the container in which the reaction takes place, the equilibrium will shift toward the left:



The opposite shift will follow a volume decrease as the reaction shift towards the right.

Table 4.2 Le Châtelier principle		
Parameter	Change	Effect
Concentration	Add reactants	→
	Remove reactants	←
	Add products	←
	Remove products	→
Volume	Increase volume	More moles
	Decrease volume	Less moles
Catalyst	Add a catalyst	No effect
Temperature		
Endothermic reaction	Increase temperature	→
	Decrease temperature	←
Exothermic reaction	Increase temperature	←
	Decrease temperature	→

Sample Problem 7

For the next endothermic reaction indicate whether the reaction will shift right (→) or left (←) after the following changes:



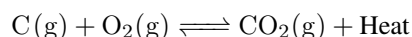
(a) adding reactants (b) adding products (c) decreasing the temperature.

SOLUTION

(a) Adding reactants always displaces the equilibrium so that reactants are consumed, hence the reaction will proceed → . (b) After adding products the reaction will tend to reduce the amount of products, and hence it will go ← . (c) The reaction is endothermic that means that it consumes heat. If we decrease the temperature it will tend to increase the temperature and hence heat needs to be formed. This will only happen if the reaction proceeds (←) .

STUDY CHECK

For next exothermic reaction indicate whether the reaction will shift right (→) or left (←) after the following changes:



(a) removing reactants (b) removing products (c) decreasing the temperature.

►Answer: (a) ← ; (b) → ; (c) → .



Experimentally, we can obtain the equilibrium concentration of the chemical species involved in an equilibrium using, for example, absorption spectroscopy. After the reactants are mixed in a given quantity, in the absence of products, the forward reaction and reverse reaction proceed until they reach a state of equilibrium, where both rates are equal. At this point, the quantity of reactants would be less than the initial mixture. This is because a small amount of these reactants will decompose and form products. Therefore, the quantity of products would increase with an increase corresponding to the amount of reactants that have decomposed. Using spectroscopy, one can measure the amount of product formed, for example, and hence elucidate the amount of reactants decomposed. One can measure the amount of reactants in equilibrium. By knowing the initial amount we mix and the resulting equilibrium concentration, we can elucidate the reaction extent. On one hand, stoichiometry plays a role in the final results of these quantities as the amounts of reactants decomposed need to form products respecting the reaction stoichiometry. On the other hand, the numerical value of the equilibrium constant K_c , characteristic of a reaction at a given temperature, is also critical. As the equilibrium concentrations of reactants and products should respect K_c . The ICE method is a tabular method used to obtain the extent of a reaction. This method leads to a mathematical function, a polynomial. Its roots give the numerical value of the extent of a reaction, and at the same time allow the calculation of the equilibrium concentration of all species involved in the chemical equilibrium.

The ICE method The ICE method is a tabular method that helps calculate the extent of a reaction for a given equilibrium. It requires the filling of a table, called the ICE table. Each column of this table will represent a reactant or a product. The table will have on its first row the chemical reaction whose equilibrium we are interested in studying. Its second row includes the initial quantities of reactants and products, in molar units. This row is called the I as Initial row. The second row, called the change, C, row, contains the unknown values of the extent of the reaction represented by the unknown variable x . The values of x should respect the reaction's stoichiometry and, for example, if the stoichiometric coefficient of a reactant is 2, the corresponding entry on the table for that reactant should be $2x$. At the same time, the entry should include negative signs for reactants and positive signs for the products. This is because reactants decompose, disappear to produce products. The last row is the equilibrium, E, row. It contains the result of adding the I and C rows. Below we have examples of ICE charts for the reaction between carbon monoxide and water to produce carbon dioxide, with initial concentrations of 0.1M and 0.2M, respectively for carbon monoxide and water

	$\text{CO}_{(\text{g})}$	+	$\text{H}_2\text{O}_{(\text{g})}$	$\rightleftharpoons$	$\text{CO}_{2(\text{g})}$	+	$\text{H}_2_{(\text{g})}$
Initial	0.1		0.2				
Change	$-x$		$-x$		x		x
Equilibrium	$0.1-x$		$0.2-x$		x		x

Another example for the decomposition of nitrosyl chloride at an initial concentration of 0.3M to produce nitrogen monoxide and chlorine

	$2 \text{NOCl}_{(\text{g})}$	$\rightleftharpoons$	$2 \text{NO}_{(\text{g})}$	+	$\text{Cl}_{2(\text{g})}$
Initial	0.3				
Change	$-2x$		$2x$		x
Equilibrium	$0.3-2x$		$2x$		x

A polynomial associated with an equilibrium The last row of the ICE chart represents the equilibrium concentration of reactants and products. Still, as we

do not know yet the extent of the reaction x we can obtain numerical values of these concentrations. Still, we know that all these concentrations are not independent of each other, and they are linked through the equilibrium constant. By plugging the equilibrium concentrations in terms of x in K_c , we can obtain a polynomial. The positive root of this polynomial—or the root that makes chemical sense—will lead to the numerical value for the extent of reaction. The order of the polynomials is related to the complexity and type of equilibrium. For example, acid-base equilibrium tends to be represented by second-order polynomials, whereas solubility equilibrium can be represented by third-degree polynomials. The use of polynomials to describe an equilibrium is not new, and this family of mathematical functions is well known in fields such as game theory, fluid mechanics, or statistical mechanics. You can use your graphing calculator to obtain the roots of any of the polynomials associated with a chemical equilibrium. Always remember only the positive root—or the root that makes chemical sense—has chemical meaning.

Concentrations at equilibrium We can easily obtain the equilibrium concentrations of reactants and products once we have obtained the positive root of the equilibrium polynomial, the extent of the reaction. Let us analyze the reaction between water and carbon monoxide to produce carbon dioxide and hydrogen given that the initial concentration of water is 0.2M and the one for carbon monoxide is 0.1M

	CO _(g)	+	H ₂ O _(g)	⇌	CO _{2(g)}	+	H _{2(g)}
Initial	0.1		0.2				
Change	$-x$		$-x$		x		x
Equilibrium	$0.1-x$		$0.2-x$		x		x

we can plug the value of the extent of reaction to calculate the equilibrium concentrations. For the example in hand, we found that the reaction extent was x , hence we have that the equilibrium concentrations, which is $K_c=5.10$ at 700K, are:

$$\frac{x^2}{(0.1-x) \cdot (0.2-x)} = K_c$$

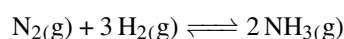
rearranging we have

$$\left(1 - \frac{1}{K_c}\right)x^2 - 0.3x + 0.02 = 0$$

Solving to x we arrive to two different solutions $x_1=0.286\text{M}$ and $x_2=0.087\text{M}$. The first answer has no chemical meaning, as we start with 0.1M of CO and 0.286M cannot decompose from 0.1M. So we have that the final reaction extent is 0.086M. Now we can go back to our ICE chart and compute the equilibrium concentration of each reactant and product: $[\text{CO}]_{Eq}=0.013\text{M}$, $[\text{H}_2\text{O}]_{Eq}=0.113\text{M}$, $[\text{CO}_2]_{Eq}=0.087\text{M}$ and $[\text{H}_2]_{Eq}=0.087\text{M}$.

Sample Problem 8

A mixture of 2 mol of N₂ and 2 mol of H₂ was placed in a 2L vessel at 720°C. At that temperature, the equilibrium constant for the formation of ammonia is $K_c=2.4 \times 10^{-3}$



Once the equilibrium is reached, calculate the equilibrium concentration of the reactants and products.

SOLUTION

We will first build up the ICE chart given that the initial concentration of



reactants is both 1M

	$\text{N}_2(\text{g})$	+	$3\text{H}_2(\text{g})$	$\rightleftharpoons$	$2\text{NH}_3(\text{g})$
Initial	1		1		
Change	$-x$		$-3x$		$2x$
Equilibrium	$1-x$		$1-3x$		$2x$

Now we will obtain the polynomial and solve for the extent. Using the expression for K_c

$$\frac{[\text{NH}_3]^2}{[\text{N}_2] \cdot [\text{H}_2]^3} = K_c$$

now substituting the values from the ICE chart:

$$\frac{(2x)^2}{(1-x) \cdot (1-3x)} = K_c$$

The corresponding polynomial will be

$$\left(\frac{4}{K_c} - 3\right)x^2 + 4x - 1 = 0$$

By solving given that $K_c = 2.4 \times 10^{-3}$, we obtain two roots, and only the positive root $x = 0.023$ has chemical sense. Now we can obtain the equilibrium concentrations for reactants and products: $[\text{N}_{2eq}] = 0.977\text{M}$, $[\text{H}_{2eq}] = 0.931\text{M}$, and $[\text{NH}_{2eq}] = 1.046\text{M}$.

◆ STUDY CHECK

Hydrogen gas reacts with molecular iodine in the gaseous state to produce hydroiodic acid, a strong, colorless, corrosive gas with acidic properties. Suppose that a mixture made of 0.1 moles of H_2 , 0.3 moles of I_2 , and 0.01 moles of HI is placed in a 0.5L flask. Given the equilibrium constant for the reaction at a given temperature



calculate the equilibrium concentration of all species.

►Answer: 0.035, 0.435, and 0.35M.



CHAPTER 4

CHEMICAL EQUILIBRIUM

4.1 True or false : (a) At equilibrium, the rate of the reverse reaction is twice the rate of the forward reaction (b) At equilibrium, the concentration of products do not change with time (c) At equilibrium, the concentration of reactants do not change with time (d) At equilibrium, the concentration of reactants and products do not change with time

4.2 True or false : (a) At equilibrium, the rate of the reverse reaction do not change (b) At equilibrium, the rate of the forward reaction do not change (c) At equilibrium, the rate of the reverse reaction equals the rate of the forward reaction (d) At equilibrium, the concentration of reactants and products are not constant

EQUILIBRIUM CONSTANTS

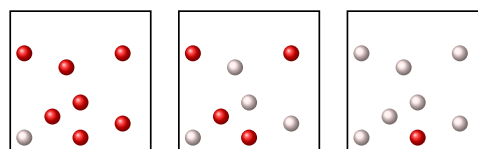
4.3 Write down the forward and reverse reactions for the following reactions in equilibrium: (a) $\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$ (b) $2 \text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{MgO}(\text{s})$

4.4 Write down the forward and reverse reactions for the following reactions in equilibrium: (a) $\text{BaCO}_3(\text{s}) \rightleftharpoons \text{Ba}^{2+}_{(\text{aq})} + \text{CO}_3^{2-}_{(\text{aq})}$ (b) $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{H}_2\text{O}(\text{l})$

4.5 For the reactions below and given the value of the equilibrium constant indicate whether the equilibrium mixture will have: (a) More reactants than products (b) More products than reactants (c) Same amount of products and reactants (a) $\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{O}_2(\text{g})$ $K_c = 0.001$ (b) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{NO}(\text{g})$ $K_c = 2 \cdot 10^{25}$ (c) $2 \text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$ $K_c = 6.4 \cdot 10^9$

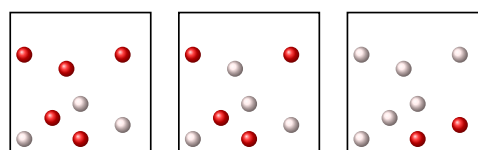
4.6 For the reactions below and given the value of the equilibrium constant indicate whether the equilibrium mixture will have: (a) More reactants than products (b) More products than reactants (c) Same amount of products and reactants (a) $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightleftharpoons 2 \text{NH}_3(\text{g})$ $K_c = 1$ (b) $2 \text{NO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2 \text{NOCl}(\text{g})$ $K_c = 6.5 \cdot 10^4$

4.7 Indicate which of the following diagrams (left, center, right) represent better the following systems at equilibrium given the K_c values (a, b, c). For example, you might say the "left diagram corresponds to for example (a)".



(a) $\text{red} \rightleftharpoons \text{grey}$ $K_c = 7$ (b) $\text{red} \rightleftharpoons \text{grey}$ $K_c = 0.1$
(c) $\text{red} \rightleftharpoons \text{grey}$ $K_c = 1$

4.8 Indicate which of the following diagrams (left, center, right) represent better the following systems at equilibrium given the K_c values (a, b, c). For example, you might say the "left diagram corresponds to for example (a)".



(a) $\text{red} \rightleftharpoons \text{grey}$ $K_c = 1$ (b) $\text{red} \rightleftharpoons \text{grey}$ $K_c = 0.6$
(c) $\text{red} \rightleftharpoons \text{grey}$ $K_c = 3$

4.9 Write down the expressions of K_c and K_p for the following reactions: (a) $2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{SO}_3(\text{g})$ (b) $\text{CO}(\text{g}) + 2 \text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{g})$ (c) $\text{C}_2\text{H}_6(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{C}_2\text{H}_5\text{Cl}(\text{s}) + \text{HCl}(\text{g})$

4.10 Write down the expression of K_c for the following reactions: (a) $\text{BaCO}_3(\text{s}) \rightleftharpoons \text{Ba}^{2+}_{(\text{aq})} + \text{CO}_3^{2-}_{(\text{aq})}$ (b) $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{H}_2\text{O}(\text{l})$

USING EQUILIBRIUM CONSTANTS

4.11 The reaction of carbon monoxide with hydrogen to produce methanol has a equilibrium constant in terms of concentration that at a certain temperature is larger than one

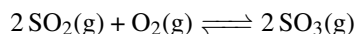


Calculate: (a) the equilibrium concentration of hydrogen (H_2) given that the equilibrium concentration of methanol (CH_3OH) and carbon monoxide (CO) for the reaction is



2M, respectively. (b) the equilibrium concentration of hydrogen (H_2) given that the equilibrium concentration of methanol (CH_3OH) and carbon monoxide (CO) for the reaction are 3M and 1M, respectively.

4.12 Consider the following reaction:



(a) Write down the expression of K_c . (b) Calculate the numerical value of K_c for the reaction if the concentrations at equilibrium at 1000K are 2M for SO_3 , 0.3M for O_2 and 1M for SO_2 . (c) indicate whether an equilibrium mixture will contain mostly products, mostly reactants or equal amounts of reactants and products.

4.13 Complete the table and calculate K_c and K_p at 300K:

	Reaction	K_c	K_p
(a)	$2 \text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3 \text{H}_2(\text{g})$	17	
(b)	$\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$		0.05

4.14 Complete the table and calculate K_c and K_p at 300K:

	Reaction	K_c	K_p
(a)	$2 \text{SO}_3(\text{g}) \rightleftharpoons 2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$	0.243	
(b)	$\text{Cl}_2(\text{g}) + \text{Br}_2(\text{g}) \rightleftharpoons 2 \text{BrCl}_2(\text{g})$		0.196

CONCENTRATION RATIO

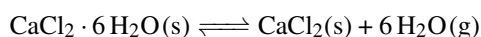
4.15 For the reactions below indicate whether they will evolve towards the right or towards the left in order to reach equilibrium.

	Reaction	K_c	Q
(a)	$2 \text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2 + 3 \text{H}_2$	17	20
(b)	$2 \text{SO}_3(\text{g}) \rightleftharpoons 2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$	0.243	10

4.16 For the reactions below indicate whether they will evolve towards the right or towards the left in order to reach equilibrium.

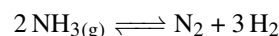
	Reaction	K_c	Q
(a)	$\text{H}_{2(\text{g})} + \text{I}_{2(\text{g})} \rightleftharpoons 2 \text{HI}_{(\text{g})}$	50	0.1
(b)	$\text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{H}_2\text{O}_{(\text{g})}$	0.196	0.196

4.17 For the decomposition of calcium chloride hexahydrate



we have that $K_c = 3.5 \times 10^{-54}$ and $Q = 10$ at 300K. Indicate towards which direction the reaction will evolve to reach equilibrium.

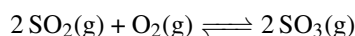
4.18 For the decomposition of calcium chloride hexahydrate



we have that $K_c = 17$ and $Q = 10$ at 300K. Indicate towards which direction the reaction will evolve to reach equilibrium.

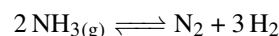
LE CHÂTELIER PRINCIPLE

4.19 Using the Le Châtelier principle indicate whether the reaction below



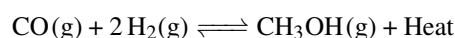
will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following actions: (a) add SO_2 (b) add SO_3 (c) remove O_2

4.20 Using the Le Châtelier principle indicate whether the reaction below



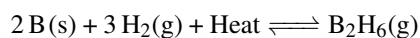
will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following actions: (a) add NH_3 (b) add N_2 (c) remove H_2

4.21 According to Le Châtelier principle indicate whether the reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after we increase temperature:

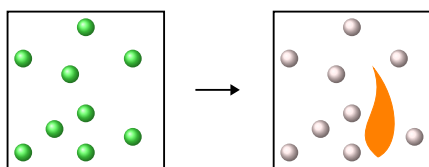




4.22 According to Le Châtelier principle indicate whether the reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after we increase temperature:

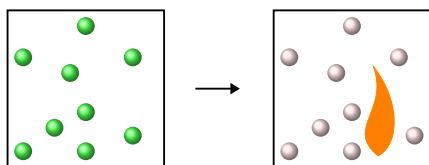


4.23 According to Le Châtelier principle indicate whether the following reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following changes:

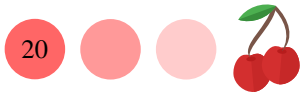


(a) adding reactants (b) increasing temperature (c) decreasing temperature

4.24 According to Le Châtelier principle indicate whether the following reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following changes:



(a) adding products (b) removing products (c) increasing temperature





Answers 4.1 (a) False (b) True (c) True (d) True 4.2 (a) True (b) True (c) True (d) False 4.3 (a) $\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$; $\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \longleftarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$ (b) $2 \text{Mg}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow 2 \text{MgO}(\text{s})$; $2 \text{Mg}(\text{s}) + \text{O}_2(\text{g}) \longleftarrow 2 \text{MgO}(\text{s})$ 4.4 (a) $\text{BaCO}_3(\text{s}) \longrightarrow \text{Ba}^{2+}_{(\text{aq})} + \text{CO}_3^{2-}_{(\text{aq})}$; $\text{BaCO}_3(\text{s}) \longleftarrow \text{Ba}^{2+}_{(\text{aq})} + \text{CO}_3^{2-}_{(\text{aq})}$ (b) $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2 \text{H}_2\text{O}(\text{l})$; $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longleftarrow 2 \text{H}_2\text{O}(\text{l})$ 4.5 (a) More reactants (b) More products (c) More products 4.6 (a) same amount (b) More products 4.7 (a) right (b) left (c) center 4.8 (a) Center (b) left (c) right 4.9 (a) $K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 \cdot [\text{O}_2]}$; $K_p = \frac{p_{\text{SO}_3}^2}{p_{\text{SO}_2}^2 \cdot p_{\text{O}_2}}$ (b) $K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{H}_2]^2 \cdot [\text{CO}]}$; $K_p = \frac{p_{\text{CH}_3\text{OH}}}{p_{\text{H}_2}^2 \cdot p_{\text{CO}}}$ (c) $K_c = \frac{[\text{HCl}]}{[\text{C}_2\text{H}_6] \cdot [\text{Cl}_2]}$; $K_p = \frac{p_{\text{HCl}}}{p_{\text{C}_2\text{H}_6} \cdot p_{\text{Cl}_2}}$ 4.10 (a) $[\text{Ba}^{2+}] \cdot [\text{CO}_3^{2-}]$ (b) $\frac{1}{[\text{H}_2]^2 \cdot [\text{O}_2]}$ 4.11 (a) 0.26 M (b) 0.46M 4.12 (a) $\frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 \cdot [\text{O}_2]}$ (b) 13.3 (c) mostly products 4.13 (a) $K_c = 17$; $K_p = 10288$ (b) $K_c = 2 \times 10^{-3}$; $K_p = 0.05$ 4.14 (a) $K_c = 0.243$; $K_p = 5.977$ (b) $K_c = 0.196$; $K_p = 0.196$ 4.15 (a) $<$ (b) $<$ 4.16 (a) $\longrightarrow$ (b) $\rightleftharpoons$ 4.17 $\longleftarrow$ 4.18 $\longrightarrow$ 4.19 (a) $\longrightarrow$ (b) $\longleftarrow$ (c) $\longleftarrow$ 4.20 (a) $\longrightarrow$ (b) $\longleftarrow$ (c) $\longrightarrow$ 4.21 $\longleftarrow$ 4.22 $\longrightarrow$ 4.23 (a) $\longrightarrow$ (b) $\longleftarrow$ (c) $\longrightarrow$ 4.24 (a) $\longleftarrow$ (b) $\longrightarrow$ (c) $\longrightarrow$